

Integrated Sensing and Communication for Real-Time Detection and Localization in Live Production Metro Networks

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Abstract We present an integrated SOP-based sensing framework for real-time detection and localization of fiber disturbances in a 31 km live production metro network. The ML classifier achieved 96.2% accuracy, while bidirectional measurements supporting median localization errors below 23 m, enabling controller-assisted alarms and maintenance operations. ©2026 The Author(s)

Introduction

Optical transport networks have evolved beyond their conventional role as data transmission infrastructures. Today, they support high-capacity, long-distance, and reliable data delivery. Meanwhile, optical fibers are intrinsically sensitive to polarization fluctuations, which can be exploited for distributed sensing [1]. Among various sensing techniques, state-of-polarization (SOP) has emerged as a promising method for early detection of subsea or terrestrial anomalies [2, 3]. SOP exploits the optical cables already deployed in the network, to continuously monitor the entire light-path in real-time, providing a scalable and cost-effective sensing solution for large-scale deployments [4].

Artificial intelligence (AI) has shown strong potential for improving network resilience through fiber fault classification [5–7]. Nevertheless, accurate event localization remains an open challenge that deserves further investigation. Prior works have explored vision-transformers for fiber fault classification and localization on bidirectional links [8], while [9] investigated localization using commercial transponders as a function of event speed. However, validation under real metro traffic conditions and environmental noise remains scarce. To address this gap, we propose a robust integrated sensing framework validated on a 31 km production metro ring network.

In our previous studies [10], we investigated the classification of various anomalous events and, in the case of a critical event likely impending a fiber cut, proactively restoration of traffic to a spare optical path [11]. Accurate event localization remains both challenging and operationally critical, since precise isolation of the affected fiber segment can enable a faster intervention and service recovery. This paper advances our SOP-based sensing by demonstrating event classification, localization, and seamless controller integration in a 31 km live metro transport network under substan-

tial background traffic and environmental noise to realize a more resilient, self-healing network.

Control-Plane Integration Workflow

This section elaborates on control plane operations for maintenance and rerouting. The integrated workflow of the proposed sensing framework is illustrated in Fig. 1. In this framework, the telemetry plane and the control plane operate in a closed loop. The telemetry plane continuously monitors the working path through real-time SOP measurements, $\{S_1, S_2, S_3\}$, and exposes early signatures of impending failures, while the control plane exploits this information to activate the protection path before a service interruption occurs. The AI classifier identifies anomalous or critical events, while the localizer estimates the event position by exploiting the correlation interval Δ_t between measurements and notifies the controller with the precise location coordinates. Unlike conventional reactive protection, the proposed framework allows predictive rerouting: upon detection of a critical event on the working path, it interacts with the software defined network (SDN) controller interfaces to trigger the activation of the pre-established backup path before the main connection fails, thereby reducing service downtime.

Event Detection using AI Classifier

Polarization fingerprints are extracted from real-time optical telemetry, derived from stokes parameters and preprocessed through noise filtering and normalization, following our earlier work in [12].

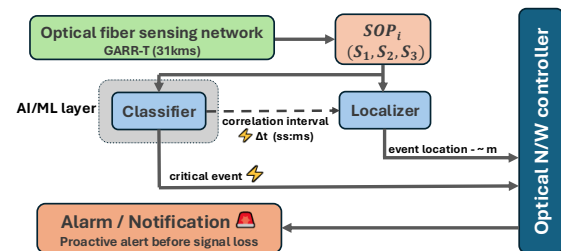


Fig. 1: Integrated workflow of the proposed sensing framework

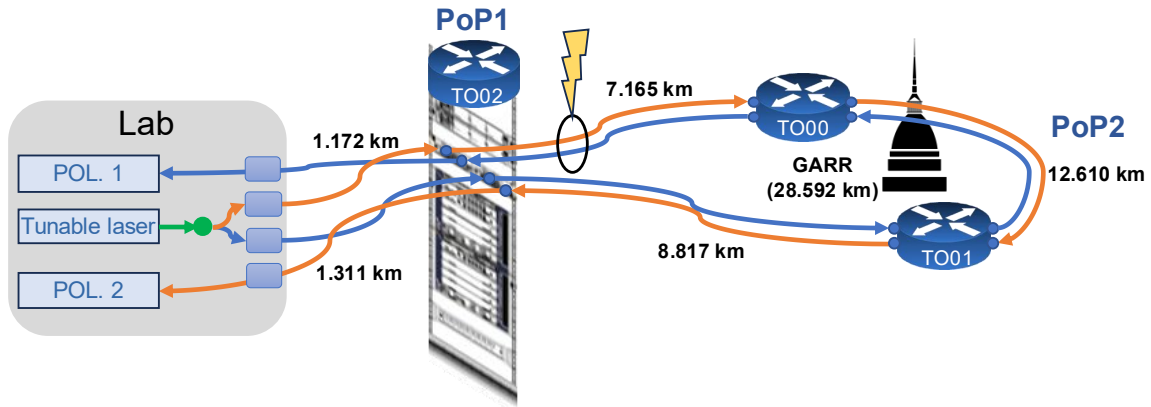


Fig. 2: Experimental setup for localization of the events in a GARR-T metro network

The vibrations were induced in the lab using an Arduino-controlled robotic arm, immediately upstream of the point of presence (PoP1), as shown in Fig. 2. The resulting mechanical perturbations were detected by a single Polarimeter (PM1000), resulting in about 2.05 million samples per event. Live metro networks are inherently noisy due to traffic, construction, railway operations, and other environmental factors, making them substantially more difficult to monitor and manage.

The data was then characterized into three distinct classes according to their vibration frequencies: 1–2 Hz events are labeled as low-frequency, 3–5 Hz as medium-frequency, and 6–10 Hz as high-frequency events. The high-frequency class is assigned a definitive label of ‘critical’, as higher vibration intensity causes the SOP to evolve more rapidly over the Poincaré sphere. The AI classifier can then trigger proactive rerouting, before physical signal degradation occurs, mitigating the potential faults [13].

In this work, we used extreme gradient boosting (XGBoost) [14], due to its low latency and robustness to noise, to analyze the anomalous polarization fingerprints extracted from the telemetry. To generalize the model, we performed feature engineering techniques, by extracting multiple statistical features from the input Stokes traces to capture both short-term and long-term trends and dependencies in the SOP data. Our model achieved an overall accuracy of 96.2%, with an average per sample inference time of only 70.52 ms, demonstrating a good balance between classifica-

tion performance and computational efficiency for real-time monitoring.

The area under the receiver operating characteristic curve (AUC-ROC) is illustrated in Fig. 3. The ROC curves show strong class separability for the low-, medium-, and high-frequency events, with all three classes achieving high true-positive rates (TPR) at very low false-positive rates (FPR). This highlights a favorable trade-off between sensitivity and specificity across different decision thresholds. The inset provides a magnified view of the low-FPR region. For the high-frequency class, the AUC value of 0.996 indicates near-perfect separability, showing that the AI classifier can reliably identify critical high-frequency events. Detected critical events are subsequently processed by the localizer module.

Localization Setup and Performance Evaluation

To evaluate the performance of the proposed localizer, we adopted a cross-correlation-based approach. The localization method was validated on a 31.07 km bidirectional loop deployed beneath the city of Turin, as illustrated in Fig. 2. The setup couples the operational GARR-T metropolitan infrastructure with the Links Foundation lab on campus, through two optical access segments of 1.172 km and 1.311 km (return), connected to PoP1. A continuous wave laser is launched at 1565.49 nm from the lab into the loop, traverses three standard single mode fiber (SSMF) spans of approximately 7.165 km, 28.592 km and 8.817 km, interconnected by two reconfigurable optical add-drop multiplexers (ROADMs), and is routed back to the lab, where two Polarimeters, Pol. 1 and Pol. 2 record the SOP and optical power traces for analysis. The blue and orange paths represent the two counter-propagating directions along the bidirectional loop.

As the fiber spans the underground metro plant, the measurements are inherently subject to realistic environmental perturbations, such as traffic and construction activity. For the experiment, vibrations were induced near PoP1, located 1.172 km, from the lab, so that the same event was observed

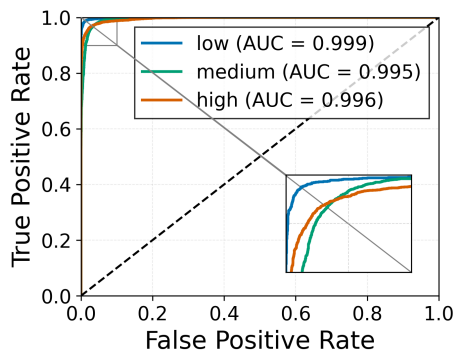


Fig. 3: AUC-ROC curves for three classes

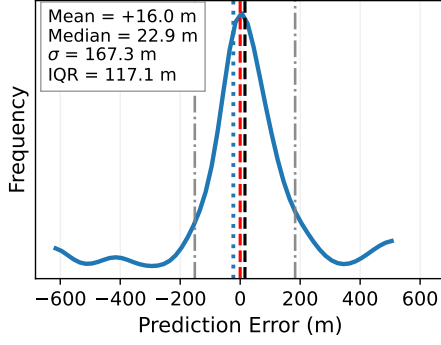


Fig. 4: Distribution of localization error relative to the true event position.

from both clockwise (CW) and counter-clockwise (CCW) directions, along the bidirectional loop. The resulting SOP transients were captured by the two Polarimeters. The two signals were synchronized using network time protocol (NTP), which provided a common time reference for the bidirectional measurements. The recorded timestamps were first used to identify the event occurrence and define the corresponding analysis window in the SOP traces. A refined estimate of the relative delay, Δt , was then obtained by applying cross-correlation to the bidirectional measurements, considering all nine combinations of the stokes parameters. To improve robustness, only correlation peaks with magnitudes above 30% of the strongest peak were retained, and the dominant peak was selected for delay estimation. Unlike threshold-based timestamping, this cross-correlation-based refinement exploits the full SOP trace and is therefore less sensitive to trigger selection and timing jitter, particularly under noisy live-network conditions.

The event location was then estimated from the time difference of arrival (TDOA) between the two measurements, as given by

$$d = \frac{L}{2} - \frac{c \Delta t}{2n}, \quad d_{\text{CCW}} = L - d_{\text{CW}} \quad (1)$$

where Δt is the cross-correlation peak lag, L is the total loop length, n is the fiber refractive index, and c is the speed of light. The second equation expresses the event location in the CCW direction as the complementary distance to the CW estimate along the fiber loop.

Random acquisitions were externally triggered to ensure precise synchronization, and the recorded stokes traces were processed offline [9]. Localization accuracy was then evaluated statistically with respect to the ground-truth location.

Fig. 4 shows the distribution of the prediction error for localization. It can be observed that most samples are concentrated around zero with a noticeable spread. The underlying statistics reveal a slight positive bias, with a mean error of +16.0 m and a median error of 22.9 m, likely attributed to hardware limitations such as timestamp quantiza-

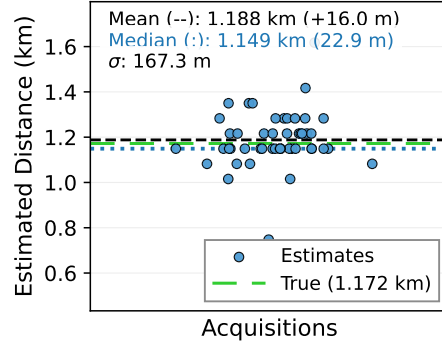


Fig. 5: Estimated event locations across multiple acquisitions

tion and finite temporal resolution. Notably, the inter-quartile range (IQR), which reflects the dispersion of the central 50% of the estimates, is 117.1 m. Despite the presence of external noise in the live metro network, the cross-correlation approach remains stable, as half of the acquisitions fall within a relatively narrow error window.

We also tested our technique by assessing the accuracy and consistency, and plotted the absolute estimated distances for all individual experimental acquisitions, as shown in Fig. 5. The true physical location of the induced disturbance was at 1.172 km from the observation point (Lab). The scatter plot illustrates location estimates (blue circles), which cluster around the ground-truth value, with a mean estimated distance of 1.188 km and median of 1.149 km, closely matching the true distance. The vertical spread of the estimates, quantified by a standard deviation (σ) of 167.3 m, reflects the measurement-to-measurement variability inherent in a live, noisy metro network, where background environmental vibrations and traffic noise can induce jitter in the cross-correlation peak.

These results affirm our method's robustness for sub-kilometer event localization in live, noisy metro networks, achieving median errors below 23 m across diverse conditions. This computational efficiency (offline processing under 1 s per trace) positions it ideally for near real-time integration with optical control planes, paving the way for automated maintenance operations.

Conclusion

The proposed SOP-based sensing framework enables real-time detection and localization of fiber disturbances in live metro networks, achieving 96.2% classification accuracy and median localization errors below 23 m under realistic operating conditions. Its integration with the control plane supports proactive fault detection and reduces service disruption. Future work will extend the framework with larger real-world datasets to further enhance localization accuracy and enable scalable deployment across wide-area fiber networks.

Acknowledgements

This work has received funding from the project PNRR-NGEU (MUR-DM117/2023), EU Horizon 2020 project ALLEGRO (GA: 101092766), SENSEI (GA: 101189545) and EWOC (GA: 101073265).

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